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Project Idea Details

Project Title

A Comparative Evaluation of the Use of Artificial Intelligence and Traditional Methods to Support Threat Detection and Incident Investigation in a Simulated Security Environment

Project Description

This project's goal is to investigate whether Artificial Intelligence tools improve threat detection and incident investigation compared with traditional security operations techniques. The problem domain is cyber security monitoring where security analysts are required to monitor logs, read and interpret alerts and respond to incidents that are of high priority all in an accurate and precision-like manner. A simulated security operations environment will be created using security information and event management/log analysis tools, sample attack datasets and AI-assisted analysis tools. The project will ask: Does AI assistance improve investigation speed, detection accuracy and explanation quality? What limitations, such as false positives and over-reliance, remain when compared with manual analyst-led investigations?

Project Tasks and Deliverables

Key Tasks

- Conduct a literature review on security operations, threat detection, incident investigation and AI-assisted cyber security.
- Design and build a simulated cyber security monitoring environment.
- Collect, generate or source safe and malicious log data.

- Create a range of controlled cyber-attack scenarios.
- Investigate each scenario using traditional/manual security methods.
- Investigate the same scenarios using AI-assisted tools and prompts.
- Compare results using metrics such as:
 - Investigation speed
 - Detection accuracy
 - False positives
 - False negatives
 - Quality of incident explanations
- Analyse findings and evaluate the effectiveness of AI-assisted approaches compared with traditional methods.

Development Work

- Configure monitoring and log analysis tools.
- Develop scripts to collect, parse or prepare datasets.
- Design and implement an evaluation framework.
- Produce result tables, graphs and visualisations for analysis.

Deliverables

- a) Simulated cyber security environment.
- b) Threat detection and investigation scenarios/datasets.
- c) Comparison framework and evaluation metrics.
- d) Results and performance analysis.
- e) Tables, charts and visualisations.
- f) Final project report.

Expected Supervisor Information

The idea was inspired by the section on Canva ‘Security: IDS and CTF’, particularly the areas of Network-based Intruder Detection Systems, Host Based Intruder Detection Systems and Security Information and Event Management. It also touches on other related topics such as Machine Learning/Deep Learning, Cyber Security issues, Ethical Hacking and Cloud security because the project considers artificial intelligence-assisted cyber security analysis.

Supervisor Name

TBD

Original Canvas Title

Security: IDS and CTF